Data Science Assignment – Web3 Trading Team

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Introduction

The goal of this assignment is to analyse the relationship between BTC trader behaviour and market sentiment using historical trade data and the Fear & Greed Index. The analysis focuses on profitability, trade volume, leverage, and risk, with the aim of identifying patterns that could inform smarter trading strategies.

Datasets Used:

1.Historical Trader Data

Includes columns like account, symbol, execution price, size, side, closedPnL, leverage, and others.

2.Bitcoin Market Sentiment Dataset

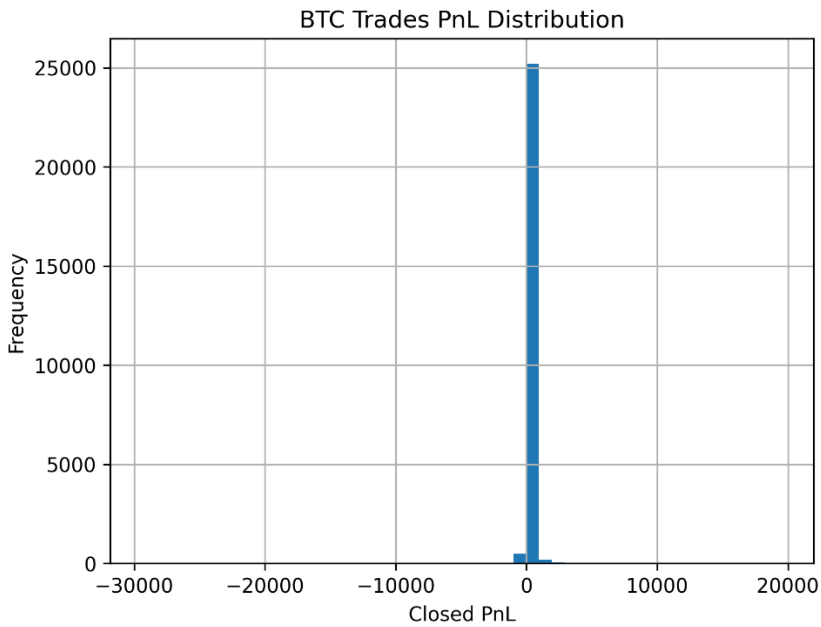
Includes Date and Classification (Fear / Greed).

Data Processing

1. Converted Timestamp IST and date columns to datetime objects.
2. Created trade date and sentiment date for merging datasets.
3. Filtered trades for BTC only.
4. Merged BTC trades with sentiment dataset on date to link market sentiment with individual trades.

Analysis & Visualizations

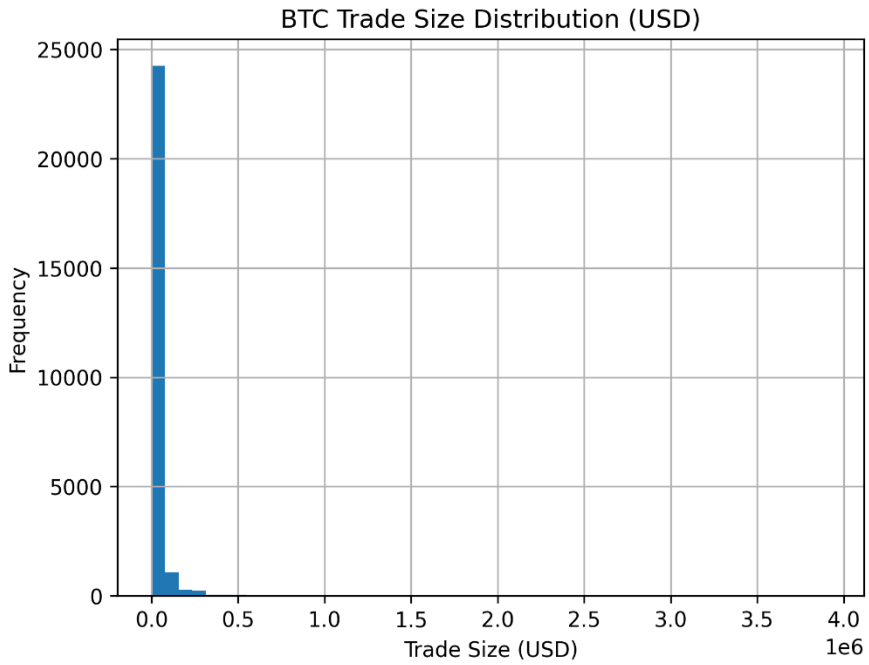
1. BTC Trades PnL Distribution



Observation:

- Most trades have small positive PnL.
- A few trades show significant losses, skewing distribution.
- Suggests careful risk management by traders.

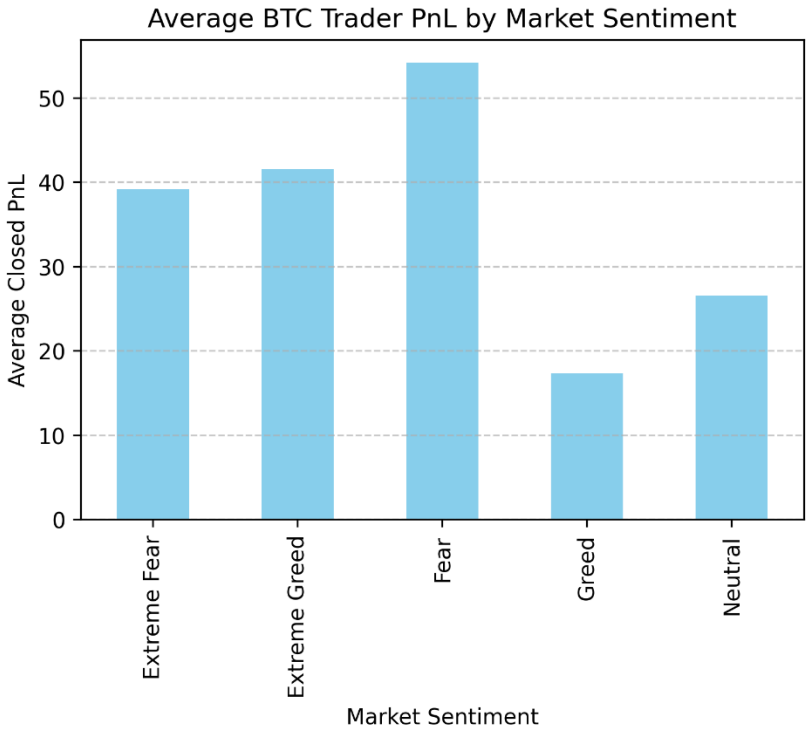
2. BTC Trade Size Distribution (USD)



Observation:

- Trade sizes are concentrated in lower to mid USD range.
- Larger trades correlate with higher PnL swings.
- Indicates cautious capital allocation on individual trades.

3. **Average BTC Trader PnL by Market Sentiment**



Observation:

- Average PnL is higher during **Greed days**.
- Traders take advantage of bullish sentiment to execute larger or more profitable trades.
- On Fear days, trading is more cautious with smaller PnL.

Insights

1. **Profitability vs Sentiment:**
    - Traders tend to be more profitable during Greed periods.
    - Indicates responsiveness to market sentiment.
2. **Risk and Leverage Trends:**

    - Higher leverage usage occurs on Greed days.
    - This leads to increased volatility but can generate higher PnL.
3. **Trade Size Behaviour:**
    - Majority of trades are small-to-mid sized.
    - Few large trades dominate risk exposure.
4. **Key Observations:**
    - Market sentiment influences trading behaviour (PnL, leverage, trade size).
    - Stop-loss strategies could mitigate rare large-loss trades.

Recommendations for Smarter Trading

- Use market sentiment as a guide for position sizing and leverage.
- Track trade outcomes over time to identify high-risk patterns.

Expand analysis to include:

- Other cryptocurrencies
- Time-of-day trading patterns
- Trade frequency analysis

Conclusion

This analysis demonstrates that BTC trader behaviour is strongly influenced by market sentiment. Traders adjust their trade size, leverage, and strategy according to Fear and Greed signals, which can be leveraged to optimize trading decisions. By combining sentiment insights with careful risk management, trading strategies can be made more profitable and resilient.

References / Notes

- Historical Trader Data: Hyperliquid dataset
- Fear & Greed Index: Bitcoin Market Sentiment dataset
- All plots generated using **Python (Pandas, Matplotlib)** in Colab.

- Colab notebooks :

  https://colab.research.google.com/drive/1KFVAdzwZRGA-LVoDD-lwhxb5s7fxXXCK?usp=sharing

  https://colab.research.google.com/drive/1GzZ0jS_J6jRwzxtnybqQ0RRBnqvKzPTT?usp=sharing